Early mineral entombment as a taphonomic confound in the post-Great Oxidation Event microfossil record

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Abstract

The diversification of morphologically diagnostic microfossils after the Great Oxidation Event (GOE; ∼2.4 Ga) is usually read as a biological signal, yet the anoxic, sulfate-poor oceans of the Archaean should, in principle, have retarded the decay of organic-walled cells and favoured their preservation. We advance the hypothesis that early mineral entombment, predominantly silicification with which early iron mineralisation is associated, is a significant taphonomic confound in this pattern, and evaluate it against the published petrographic and geochemical record of seven chert-hosted biotas spanning the GOE, including the 2.45–2.21 Ga Turee Creek Group, which straddles the boundary. Within the 1.88 Ga Gunflint biota, morphospecies that retain internal cellular structure carry 10^8–10^9 intracellular Fe atoms μm^{-3}, two to three orders of magnitude above empty sheaths, indicating that cellular fidelity is linked to early mineralisation; this iron may, however, reflect in vivo biomineralisation rather than a purely external agent, so the observation supports the principle that fidelity requires mineralisation before decay without isolating silica as the sole mechanism. Across the seven biotas, fidelity tracks independent petrographic indicators of early silica cementation rather than any measure of oxygenation, and the silica cycle did not change across the GOE. The biomarker evidence once advanced for pre-GOE eukaryotes is best regarded as contaminant, while the obligate aerobicity of sterol biosynthesis gives an independent, non-taphonomic reason for crown-eukaryote diversification to track oxygen. The post-GOE "rise" of microfossils therefore plausibly reflects, in part, a facies-controlled preservational signal, but the partition between evolution and unmasking remains unresolved and is lineage-conditional; it cannot be settled from the rock record alone.

Keywords: Great Oxidation Event; taphonomy; silicification; iron mineralisation; Archaean–Proterozoic; microfossils; molecular clocks; sterol biosynthesis.

1 Introduction

It has long been recognised that anoxia retards the degradation of sedimentary organic matter, and that the destruction of cellular morphology during early decay proceeds more slowly where molecular oxygen is scarce (Canfield, 1994; Hedges & Keil, 1995). From this premise one might expect the Archaean ocean, in which dissolved oxygen was negligible (Pavlov & Kasting, 2002) and sulfate scarce (Habicht et al., 2002; Crowe et al., 2014), to have been a favourable setting for the preservation of organic-walled microfossils. The rock record appears to say the opposite. Mat-related carbonaceous remains are known from several Archaean successions, but assemblages of taxonomically diagnostic microfossils become abundant and widespread only after the GOE (Javaux & Lepot, 2018; Wacey et al., 2014), as exemplified by the gunflintiacean biotas of the ∼1.88 Ga Gunflint Iron Formation (Lepot et al., 2017).

This contrast is open to two readings. The first is biological: oxygenic photosynthesis, or the eukaryotic cell, radiated after the GOE, and the record documents that diversification. The second is preservational: the conditions required to preserve morphology became more widely available, and pre-existing diversity became visible. The two are not mutually exclusive, but the distinction is consequential. If the latter contributes substantially, first-appearance data cannot be taken at face value as a record of origination.

This paper does not present new measurements. It synthesises the published petrographic, spectroscopic and isotopic evidence for seven chert-hosted biotas on either side of the GOE, and evaluates the hypothesis that early mineral entombment is a sufficient taphonomic confound to qualify a purely biological reading of the post-GOE pattern. The synthesis is offered as a hypothesis, not as a settled result: the supporting data are semi-quantitative, and several of the central observations are themselves open to alternative interpretation, as the discussion makes explicit.

2 Background

2.1 Redox across the GOE

The disappearance of mass-independent sulfur isotope fractionation between ∼2.45 and 2.32 Ga marks the GOE as the interval in which atmospheric O_2 rose irreversibly above $\sim 10^{-5}$ of the present atmospheric level (Holland, 2006; Bekker et al., 2004; Pavlov & Kasting, 2002; Farquhar et al., 2000). The transition was protracted and regionally diachronous (Lyons et al., 2014; Philippot et al., 2018). Iron speciation indicates that anoxic, ferruginous conditions dominated deposition throughout the Archaean and much of the Proterozoic (Canfield, 1998; Poulton & Canfield, 2005, 2011), euxinia being spatially restricted (Reinhard et al., 2009). Seawater sulfate rose across the GOE from micromolar to millimolar levels (Habicht et al., 2002; Kah et al., 2004; Crowe et al., 2014), but sulfur isotope evidence of this rise extends into the Neoarchean (Reinhard et al., 2009), and signals of oxidative weathering precede the GOE in several basins (Anbar et al., 2007).

2.2 The microfossil record and the Turee Creek Group

The Archaean record of cellular morphology is sparse and contested. The filamentous structures described from the ∼3.46 Ga Apex chert (Schopf, 1993) were reinterpreted as abiogenic (Brasier et al., 2002), and although better-preserved Archaean microbiotas are known from the ∼3.4 Ga Strelley Pool Formation (Sugitani et al., 2010; Delarue et al., 2020), their cellular fidelity is modest in comparison with later assemblages. The most directly relevant succession is the Turee Creek Group (∼2.45–2.21 Ga), which spans the GOE itself and preserves diverse microfossil communities in black chert (Barlow & Van Kranendonk, 2018), the taphonomy and iron mineralisation of which have been documented by Lepot et al. (2017a). Gunflint-type biotas of ∼1.9 Ga age carry filaments, spheroids and spindle-shaped forms in cellular detail (Lepot et al., 2017; Javaux & Lepot, 2018).

2.3 The silica cycle

Before the evolution of silica-biomineralising organisms, the ocean was saturated or supersaturated with respect to amorphous silica, with dissolved concentrations of order ∼2 mM and abiological precipitation the dominant sink (Siever, 1992; Perry & Lefticariu, 2007); early silicification is implicated in exceptional preservation into the Neoproterozoic (Tarhan et al., 2016). The drawdown of seawater silica was driven by the much later radiation of sponges, radiolaria and diatoms. The capacity for early silica precipitation was therefore available on both sides of the GOE.

2.4 An independent, biochemical oxygen constraint

Sterol biosynthesis is obligately aerobic: the cyclisation and oxidative modification of squalene to yield sterols requires molecular oxygen, on the order of eleven molecules of O_2 per sterol (Gold et al., 2017; Desmond & Gribaldo, 2009). Crown eukaryotes synthesise sterols, and the last eukaryotic common ancestor is inferred to have done so (Desmond & Gribaldo, 2009). Irrespective of preservation, this provides a non-taphonomic reason to expect crown-eukaryote diversification to track the availability of oxygen.

Table 1: Preservation of cellular morphology in seven chert-hosted biotas, with the petrographic criterion used to assess the timing of silicification, kept separate from the fidelity outcome.

Unit (age)	GOE	Silicification criterion (independent of fidelity)	Preservation of morphology	Source
Strelley Pool (∼3.4 Ga)	pre	early silicification; wall organic matter partly replaced by microquartz/chalcedony	diverse (lenticular, spinose) but lower cellular fidelity than Gunflint	(Sugitani et al., 2010; Delarue et al., 2020; Wacey et al., 2012)
Moodies Gp (∼3.22 Ga)	pre	silicification of mat textures only; cells not encapsulated	tufted, cavity-dwelling mats; cellular detail poorly resolved	(Homann et al., 2015)
Tumbiana Fm (∼2.72 Ga)	pre	organic matter hosted in carbonate; chert present but cells not silica-encapsulated	kerogenous globules; rare filaments; no cellular assemblage	(Lepot et al., 2009; Decraene et al., 2021)
Turee Creek Gp (∼2.45–2.21 Ga)	spans	early chert permineralisation with associated Fe mineralisation	diverse spheroids and filaments across the GOE	(Lepot et al., 2017a; Barlow & Van Kranendonk, 2018)
Belcher Gp (∼2.0 Ga)	post	chert replacement of stromatolitic carbonate	diverse microbiota, chert-permineralised	(Javaux & Lepot, 2018)
Gunflint Fm (∼1.88 Ga)	post	pre-compaction silica encapsulation; intra-microfossil microquartz finer than matrix	diverse, high cellular fidelity; intra-cellular Fe-silicate/carbonate	(Lepot et al., 2017; Wacey et al., 2013; Alleon et al., 2016)
Sokoman IF (∼1.88 Ga)	post	early chert of iron-formation	Gunflint-type assemblage	(Javaux & Lepot, 2018)

3 Material and approach

Seven chert-hosted biotas were compared (Table 1): the Strelley Pool Formation, the Moodies Group and the Tumbiana Formation (pre-GOE), the Turee Creek Group (which spans the GOE), and the Belcher Group, the Gunflint Iron Formation and the Sokoman Iron Formation (post-GOE). These units span shallow-marine to peritidal silicified facies of comparable, low metamorphic grade, holding thermal maturity approximately constant. For each unit the published descriptions of petrography, Raman spectroscopy, iron mineralogy and sulfur geochemistry were compiled; no new measurements are reported, and where published constraints are absent this is stated rather than inferred.

To reduce the circularity inherent in grading preservation and silicification from the same descriptive literature, the degree of early silicification is assessed from criteria independent of the preservation outcome being explained: the timing of silica cementation relative to compaction fabrics (pre-compaction, wall-conforming microquartz or chalcedony indicating early entombment, versus late megaquartz void-fill), and the quartz cement fabric. These criteria, and the unit to which each applies, are given in Table 1.

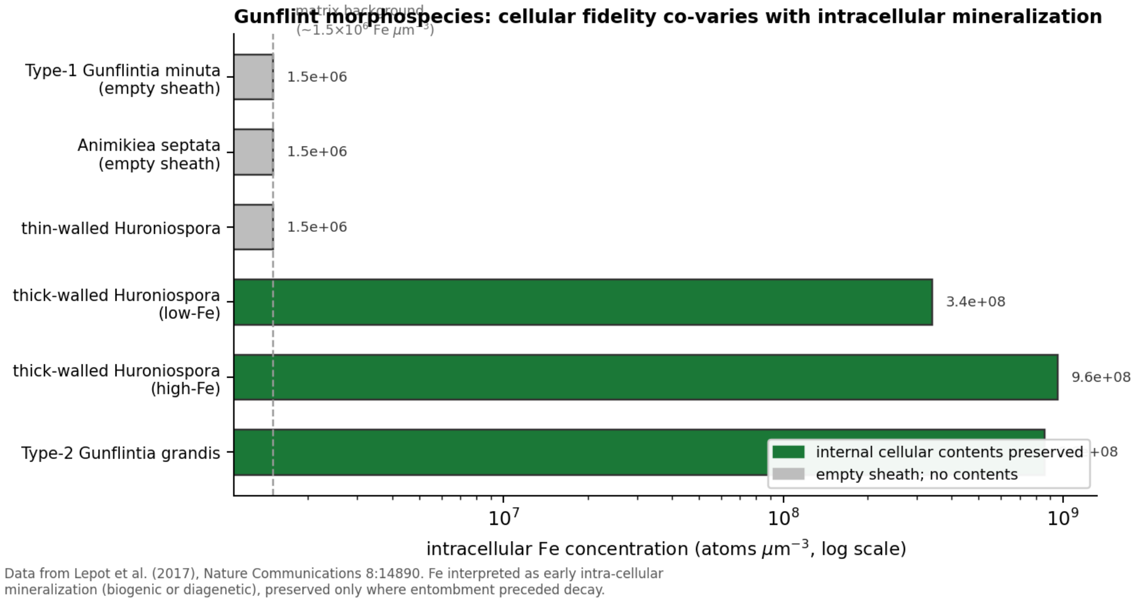

Figure 1: Intracellular Fe concentration in Gunflint morphospecies, against the preservation of cellular contents. Morphospecies with preserved internal structure carry 10^8–10^9 Fe atoms μm^{-3}; empty sheaths remain at the matrix background. The iron may reflect in vivo biomineralisation (Lepot et al., 2017), so the covariation links fidelity to early mineralisation without isolating silica as the agent. Data from Lepot et al. (2017).

4 Evidence

4.1 A within-assemblage covariation, and its interpretation

At the scale of individual populations, the Gunflint biota shows that the preservation of internal cellular structure is coupled to early mineralisation. Morphospecies retaining cellular contents, including thick-walled *Huroniospora* and Type-2 *Gunflintia*, carry between 3×10^8 and 10^9 intracellular Fe atoms μm^{-3}, together with intra-microfossil quartz finer than the surrounding matrix; empty sheaths remain at the chert background ($\sim 1.5 \times 10^6$ atoms μm^{-3}; Fig. 1) (Lepot et al., 2017).

This correlation carries a caveat that bears on the argument. Lepot et al. (2017) interpret the intracellular iron as consistent with in vivo biomineralisation by the organisms themselves, plausibly oxygenic photosynthesisers, rather than as a purely external, abiotic precipitate. If so, the mineral phase associated with fidelity is partly a metabolic product, and the clean opposition between preservation and biology is less clear-cut than a taphonomic reading would wish: the biology of the cells and their preservation are entangled. The observation nevertheless establishes the principle that morphological fidelity required mineralisation before decay had destroyed internal structure. It does not, by itself, identify silica as the agent. The assignment of silicification as the entombment medium rests instead on silica-specific evidence: the three-dimensional preservation of Gunflint cells within quartz, which required silica encapsulation and impregnation before compaction (Lepot et al., 2017; Alleon et al., 2016), and the wall-conforming microquartz documented in silicified biotas more generally (Wacey et al., 2012). We therefore frame the mechanism as early mineral entombment, of which silicification is the dominant medium in chert-hosted assemblages and with which early iron mineralisation, whether biogenic or diagenetic, is associated.

4.2 The between-assemblage pattern

Across the seven units, the preservation of cellular morphology tracks the independent indicators of early silicification, not the position of a unit relative to the GOE (Table 1). The pre-GOE Strelley Pool biota, where silicification was early, preserves a diverse assemblage, albeit with lower cellular fidelity than Gunflint and with wall organic matter partly replaced by silica (Wacey et al., 2012). The pre-GOE Tumbiana and Moodies biotas, in which silicification was confined to mat textures or absent at the cellular scale, preserve kerogen and mats rather than cells (Lepot et al., 2009; Homann et al., 2015). The Turee Creek Group, which spans the GOE, preserves diverse silicified microfossils with associated iron mineralisation (Lepot et al., 2017a; Barlow & Van Kranendonk, 2018), a pattern that is continuous across the boundary rather than stepped at it. The post-GOE Gunflint, Sokoman and Belcher biotas are early-silicified and morphologically rich (Lepot et al., 2017; Javaux & Lepot, 2018).

Pyrite is a poor indicator of fidelity. In Gunflint, pyritisation is patchy, develops under locally sulfate-limited conditions, and destroys rather than preserves organic structure (Wacey et al., 2013; Lepot et al., 2017); it is also documented in pre-GOE Tumbiana stromatolites (Decraene et al., 2021), and so cannot mark a post-GOE threshold.

4.3 Why redox and the silica cycle do not supply a GOE control

If a change in redox across the GOE had governed preservation, the proxy record of that change should align with the palaeontological pattern. It does not. Iron speciation indicates pervasive ferruginous deposition on both sides of the GOE (Canfield, 1998; Poulton & Canfield, 2011), and the sulfur isotope signal of rising sulfate extends into the Neoarchean (Reinhard et al., 2009; Philippot et al., 2018). The sulfate rise registered by δ^{34}S bears on microbial sulfate reduction and the sulphurisation of organic matter, a pathway that affects the survival of bulk kerogen rather than the templating of cellular morphology. The capacity for early silica precipitation was likewise unchanging across the GOE: Precambrian seawater remained silica-saturated, and the silica cycle was not reorganised until the much later appearance of biosilicifiers (Siever, 1992; Perry & Lefticariu, 2007).

5 Origins, first appearances, and the sterol constraint

If the post-GOE increase in morphological diversity reflects, even in part, the unmasking of lineages that existed but were not previously preserved, independent temporal evidence should place those lineages before the GOE. Molecular clocks place the stem origin of cyanobacteria, and with it oxygenic photosynthesis, in the Archaean (Fournier et al., 2021), whereas the crown-group diversification of extant Oxyphotobacteria postdates the rise of oxygen (Shih et al., 2017). Molecular clock estimates for the last eukaryotic common ancestor remain model-dependent, but several precede the first widely accepted eukaryotic microfossils, leaving a protracted gap between molecular and palaeontological estimates (Betts et al., 2018; Eme et al., 2014) (Fig. 2).

The biomarker record cannot at present resolve this gap. The steranes and 2-methylhopanes reported from $\sim$2.7 Ga shales, once taken as evidence for Archaean eukaryotes and cyanobacteria (Brocks et al., 1999), were shown to reside in younger carbonate veins (Rasmussen et al., 2008), and analysis of ultraclean drill cores has shown hopane and sterane concentrations in Archaean rocks to be indistinguishable from procedural blanks (French et al., 2015). The molecular fossil evidence for pre-GOE eukaryotes and oxygenic photosynthesis is therefore best regarded as contaminant.

A separate, biochemical consideration cuts across a purely preservational reading for eukaryotes. As noted above, sterol biosynthesis is obligately aerobic (Gold et al., 2017; Desmond & Gribaldo, 2009), so crown eukaryotes have an independent reason to diversify once oxygen became available, regardless of preservation. The clock-to-fossil gap for eukaryotes is thus not necessarily a preservational lag; it may in part reflect genuine, oxygen-limited late diversification. The unmasking hypothesis is consequently better supported for cyanobacteria, whose metabolism clocks to the Archaean, than for sterol-synthesising eukaryotes. Any reading of the post-GOE pattern must therefore be lineage-conditional.

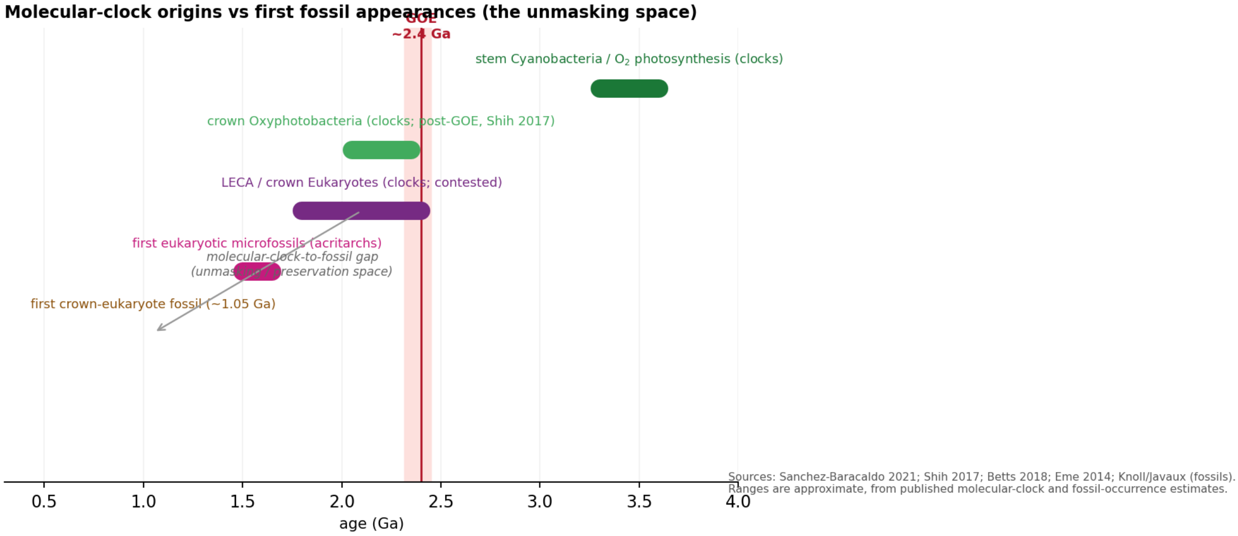

Figure 2: Molecular clock estimates of lineage origin against first fossil appearances. The stem origin of cyanobacteria lies in the Archaean (Fournier et al., 2021); crown Oxyphotobacteria diversify around or after the GOE (Shih et al., 2017); eukaryote molecular-clock estimates precede the first widely accepted eukaryotic fossils (Betts et al., 2018; Eme et al., 2014). Ranges are approximate.

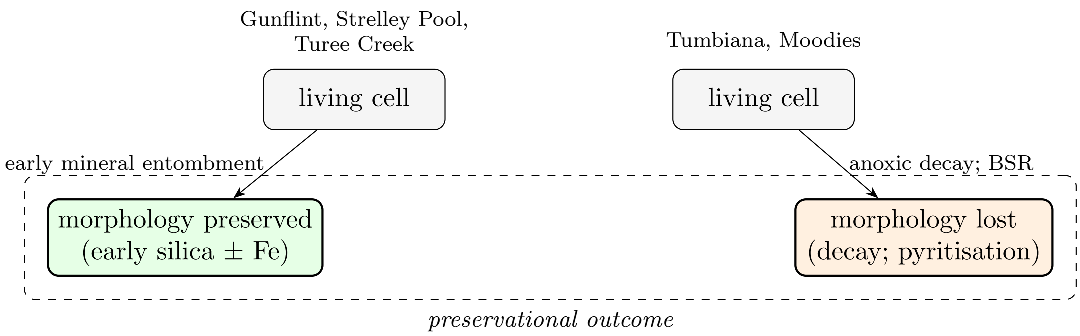

Figure 3: Preservational pathways. Where early mineral entombment (silica, with associated iron mineralisation that may be biogenic) outpaces decay, morphology is retained; where decay, and pyritisation, precede mineralisation, morphology is lost and kerogen survives. The balance is set by depositional facies and the timing of cementation.

6 Discussion

The compiled evidence is consistent with the hypothesis that early mineral entombment is a significant confound in the post-GOE microfossil record, but it does not warrant the stronger claim that oxygenation was irrelevant to preservation. The defensible position is narrower. The preservation of cellular morphology in Precambrian cherts is conditioned by early mineralisation, predominantly silicification, that operated throughout the Archaean and Proterozoic but was expressed unevenly, depending on the silica budget of the depositional environment and the timing of cementation relative to decay. Where entombment was early and complete, as at Gunflint and, to a lesser degree, Strelley Pool and the Turee Creek Group, morphology survives in cellular detail; where it was weak or confined to mat textures, as at Tumbiana and Moodies, only kerogen and mats remain (Fig. 3). Oxygenation, and the sulfate rise that accompanied it, plausibly modified the decay of organic matter and the sulphurisation of kerogen (Lepot et al., 2009; Decraene et al., 2021) but did not gate the silicification on which cellular fidelity depends.

Three implications follow, each qualified. First, abundance and diversity counts compiled across the GOE cannot be read as proxies for biomass or evolutionary innovation without correction for the silicification available in the sampled facies; the intensively sampled gunflint-type cherts are liable to over-represent the biology of their time relative to less completely silicified successions. Second, the gap between molecular clock estimates of origination and first appearance, most conspicuously for eukaryotes (Betts et al., 2018), is consistent with a preservational lag, but for sterol-synthesising lineages it may equally reflect oxygen-limited diversification; the two cannot be distinguished from the rock record alone. Third, for the search for biosignatures in ancient successions, on Mars as on Earth, the relevant variable for the preservation of morphology is a silica-precipitating facies, not the presence of oxygen.

The limitations are substantial and are stated plainly. The comparison rests on a synthesis of published descriptions rather than a uniform new dataset. The within-Gunflint relationship between intracellular iron and fidelity is quantitative but is expressed in iron, not silica, and the iron may be biogenic; the assignment of silicification as the entombment medium rests on qualitative petrography. The between-formation pattern is semi-quantitative, and the ranking of silicification, though based on criteria independent of fidelity, is drawn from a heterogeneous literature. A standardised, per-population scoring of silicification and fidelity on matched samples, regressed within single formations of comparable thermal maturity, would convert the hypothesis into a direct test.

7 Conclusions

The hypothesis advanced here is that early mineral entombment, predominantly silicification, is a significant taphonomic confound in the apparent post-GOE diversification of microfossils. The published record is consistent with this hypothesis: within Gunflint, cellular fidelity co-varies with early mineralisation, and across seven biotas fidelity tracks independent indicators of early silica cementation rather than oxygenation, while the silica cycle did not change across the GOE. The evidence does not, however, sustain the stronger claim that oxygenation governed preservation. The intracellular iron on which the within-Gunflint correlation rests may be biogenic, and the argument for silica as the entombment medium rests on qualitative petrography; the biomarker evidence for pre-GOE eukaryotes is best regarded as contaminant; and the obligate aerobicity of sterol biosynthesis provides an independent reason for crown-eukaryote diversification to track oxygen. The partition of the post-GOE pattern between evolution and unmasking is therefore unresolved and lineage-conditional. A within-formation, quantitative test of fidelity against silicification is the appropriate next step.

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